Tushar Kumar

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Professional Summary

AI Engineer with 2+ years shipping RAG and multi-agent systems alongside enterprise C/C++. Currently **Software Engineer I** at **Samsung R&D India**, where a 3-person team built a **RAG** chatbot for module specs and automated test generation, and an interactive learning platform serving **5,000+** students. Profiling work cut memory use by **25%** on Tizen deployments across **10,000+** devices; a diagnostics rewrite cut issue-resolution time by **30–40%**. Won Samsung's **Good Idea Award** (1 of 50 globally). **B.Tech. AI & Data Science**, IIT Jodhpur.

Experience

Samsung R&D Institute India, Delhi — **Jul 2024 – Present**
Software Engineer I — *Noida, UP, India*

GenAI & RAG Systems

- Built a RAG chatbot with a team of 3 that answers spec, feature, and test-case questions for Tizen modules such as SignagePlayer, replacing scattered documents with one reference point.
- Extended it to generate test cases automatically for module features and to walk engineers through first-line troubleshooting.
- Implemented the retrieval path: semantic chunking, hybrid keyword and dense vector search, and cross-encoder reranking over isolated per-corpus namespaces.

AI Learning Platforms

- Developed an interactive educational podcast platform that produced **100+** hours of lecture content for **5,000+** students, letting students interrupt playback mid-sentence to ask questions.
- Wrote the streaming layer on full-duplex WebSockets so heavy render work never stalls audio, holding barge-in response under 200 ms.
- Shipped an adaptive short-form video pipeline that expands a topic and grade level into a sequenced lesson set, with schema-constrained LLM output feeding the database directly.
- Prototyped Code2Documentation, which won Samsung's **Good Idea Award** (1 of 50 globally).

Smart Diagnostics & Logging

- Wrote a diagnostics service that collects log, CPU, and memory telemetry from **10,000+** Samsung devices.
- Cut issue-resolution time by **30–40%**; scaled log collection to 2 GB+ per session.
- Integrated the logging pipeline into Samsung's internal portal, now used by 500+ engineers in 20+ countries.

Tizen Signage Core Modules (C/C++)

- Maintain Tizen modules (SignagePlayer, OfficeViewer, Event Manager, Remote Manager) across 100+ deployments.
- Reduced memory usage by **25%** via profiling and C/C++ optimization; resolved 50+ production crashes.
- Added subtitle support to SignagePlayer for multilingual accessibility in international deployments.

Education

Indian Institute of Technology (IIT) Jodhpur — **Jul 2020 – May 2024**
B.Tech. Artificial Intelligence & Data Science — ***CGPA: 7.26/10***

- **Coursework:** Data Structures & Algorithms, Operating Systems, Computer Networking, Database Management Systems, Computer Architecture, Machine Learning, Deep Learning, Pattern Recognition & ML, Computer Vision, NLP, System Design

Technical Skills

LLM & RAG: Retrieval-Augmented Generation (RAG), Agentic RAG, Multi-Agent Systems, Prompt Engineering, Large Language Models (LLMs), LangChain, LangGraph, Model Context Protocol (MCP), FAISS, Vector Databases, Semantic Chunking, Hybrid Search (BM25 + Dense), Cross-Encoder Reranking, Knowledge Graphs, Gemini API
Machine Learning: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, OpenCV, NLTK, Deep Learning, Computer Vision, Natural Language Processing (NLP), Transfer Learning, Model Compression, PCA/LDA/t-SNE
Languages: Python, C/C++, Rust, Go, Java, JavaScript/TypeScript, SQL
Backend & Architecture: FastAPI, Django, Flask, REST APIs, WebSockets, Async & Streaming Pipelines, Event-Driven Architecture, Microservices, System Design (HLD/LLD), SQLite, SQL
DevOps & Tools: Docker, AWS, GitHub Actions, CI/CD, Git, Linux, Neovim, LaTeX
Web & IoT: ReactJS, NextJS, NodeJS, HTML/CSS/JS, Raspberry Pi, ThingSpeak

Projects

The Cutting Room | FastAPI, React 19, Docker, Gemini, ffmpeg, Python/TypeScript **2026**

- Built a multi-agent GenAI pipeline: a FastAPI hub and React 19 dashboard, with four agents coordinating over HTTP and no shared filesystem.
- Wrote the scoring engine that blends engagement signals (rate, reach, outlier, velocity) into a 0–100 score with configurable tier thresholds.
- Containerized as a multi-arch Docker image published to GHCR, including a custom ffmpeg build stripped to ∼130 MB.
- Implemented Gemini 2.5 Pro blueprint generation with jsonschema validation and automatic repair loops for shot-by-shot content breakdowns.

PageIndex – Vectorless RAG in Rust | Rust, SQLite, async-openai, serde **2026**

- Built a retrieval engine that replaces vector databases with hierarchical ToC generation and LLM-guided traversal, using no embeddings at any stage.
- Added a multi-provider LLM abstraction layer (OpenAI, Grok, Gemini, Claude, custom) with async query execution and a SQLite-backed index; lookups run under 200 ms.
- Implemented markdown, text, PDF, and ZIP ingestion with recursive directory parsing; ships as a single-binary Rust CLI.

Facial Emotion Recognition using Deep Learning | PyTorch, ResNet-34, FER-2013, OpenCV **2025**

- Trained a ResNet-34 on FER-2013 (35,887 grayscale images, 7 emotion classes) for facial expression recognition.
- Used transfer learning from ImageNet weights with a custom classifier head; wrote the preprocessing, training, and hyperparameter-tuning pipeline.
- Added face detection so the model runs on a live webcam feed.

Electronic Nose System for Gas Sensor Data | Python, PyTorch, Raspberry Pi **Sep 2023 – Apr 2024**

- Built an ML pipeline that processes **10,000+** sensor readings to classify gases and predict concentrations, running entirely on a Raspberry Pi with ThingSpeak.
- Used PCA, LDA, and t-SNE for feature extraction, then trained classifiers for gas type and regression models for concentration levels.
- Added a dashboard for live sensor visualization and automated predictions.

Table Structure Recognition using Deep Learning | PyTorch, CNN, OpenCV **Aug 2023 – Apr 2024**

- Trained CNN models for table detection and structure recognition on benchmark document datasets.
- Improved detection accuracy and structural parsing for document understanding.
- Research conducted at IIT Jodhpur.

Neural Tangent Kernel Aware Pruning in DNN | PyTorch **Jan 2023 – May 2023**

- Proposed a pruning method based on Neural Tangent Kernel (NTK) theory that removes connections while preserving training dynamics.
- Achieved ∼20% model compression with minimal accuracy loss.
- Research conducted under Prof. Binod Kumar at IIT Jodhpur.

Toxic Comment Classification on Social Media Data (NLP) | PyTorch, NLP, Docker, Streamlit **Mar 2022**

- Trained PyTorch NLP models to detect toxic content in social media datasets, comparing several classifiers on accuracy.
- Deployed as a Docker and Streamlit web application for interactive model testing on user-provided text.

Handmade Sketch Colorization using Segmentation | Python, OpenCV, Docker, Streamlit **Jan 2022 – May 2022**

- Built a web app that colorizes hand-drawn sketches using classical segmentation techniques.
- Containerized the pipeline with Docker; deployed via Streamlit supporting 100+ concurrent sessions.

Certifications

LangChain & LangGraph – Agentic AI Engineering	Udemy, May 2026
Prompt Engineering for Everyone Bootcamp	Udemy, May 2026
The Complete MCP (Model Context Protocol) Masterclass	Udemy, May 2026
Rust: The Complete Developer's Guide	Udemy, Feb 2026
Fundamentals of Operating Systems	Udemy, Jun 2025
Mathematics for Machine Learning (PCA, Linear Algebra, Calculus)	Coursera, 2022–23